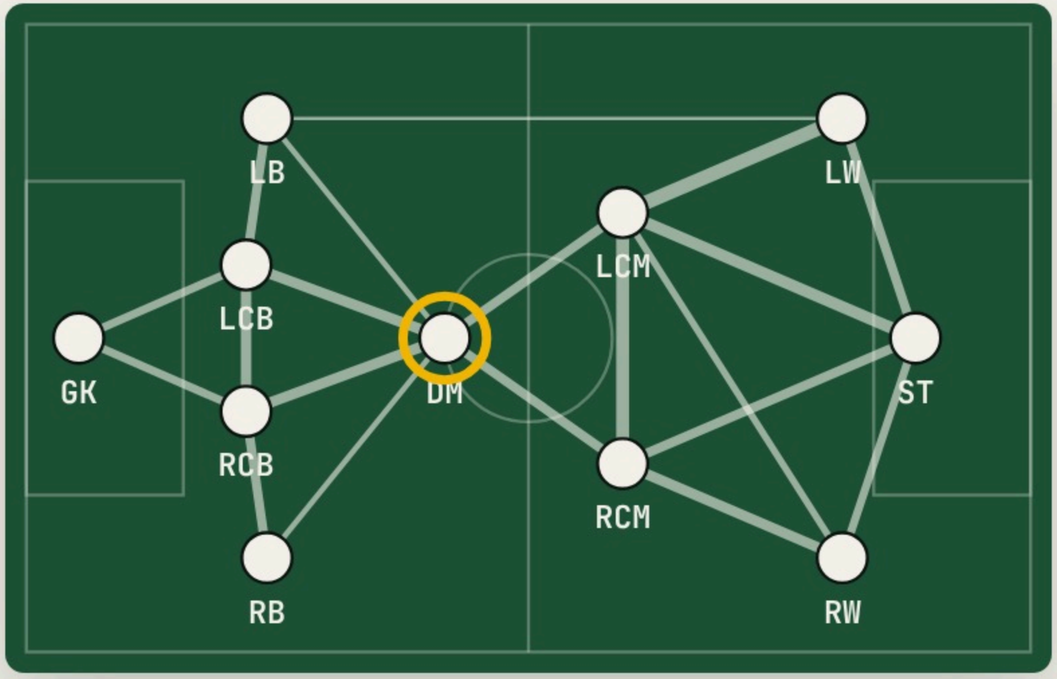
FAULT LINES · LIVE PASS-NETWORK INTELLIGENCE

EVERY TEAM HAS A FAULT LINE.

The one player its passing can't survive without. Fault Lines finds them live and tells everyone watching what breaks without them.



Illustrative network on this slide and slides 3 and 4 · live demo: harshul27.github.io/faultlines

THE PROBLEM

MATCH DATA COUNTS WHAT PLAYERS DO, NOT WHAT THE TEAM NEEDS

WHAT MATCH FEEDS COUNT

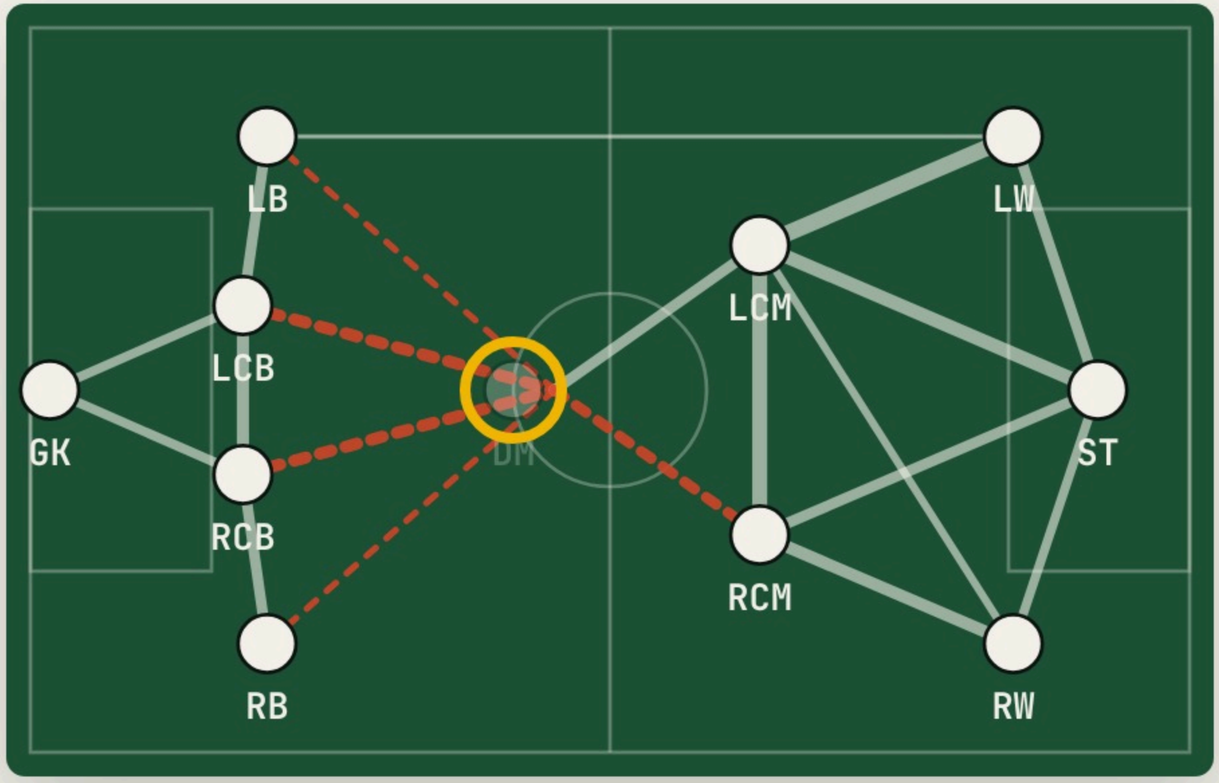
- Passes per player
- Possession share
- Expected goals
- Heatmaps

WHAT PEOPLE WATCHING ASK

- Who does this team depend on right now?
- What breaks if that player is marked or subbed?
- Is our game plan working, and who is off it?

THE INSIGHT

A BUSY PLAYER AND AN ESSENTIAL PLAYER ARE DIFFERENT PEOPLE



BUSIEST PLAYER OUT

5% of links lost

LCM, 37 passes. The rest still connect.

FAULT LINE OUT

23% of links lost

DM, 32 passes. The back line is cut off from the front.

HOW IT MEASURES

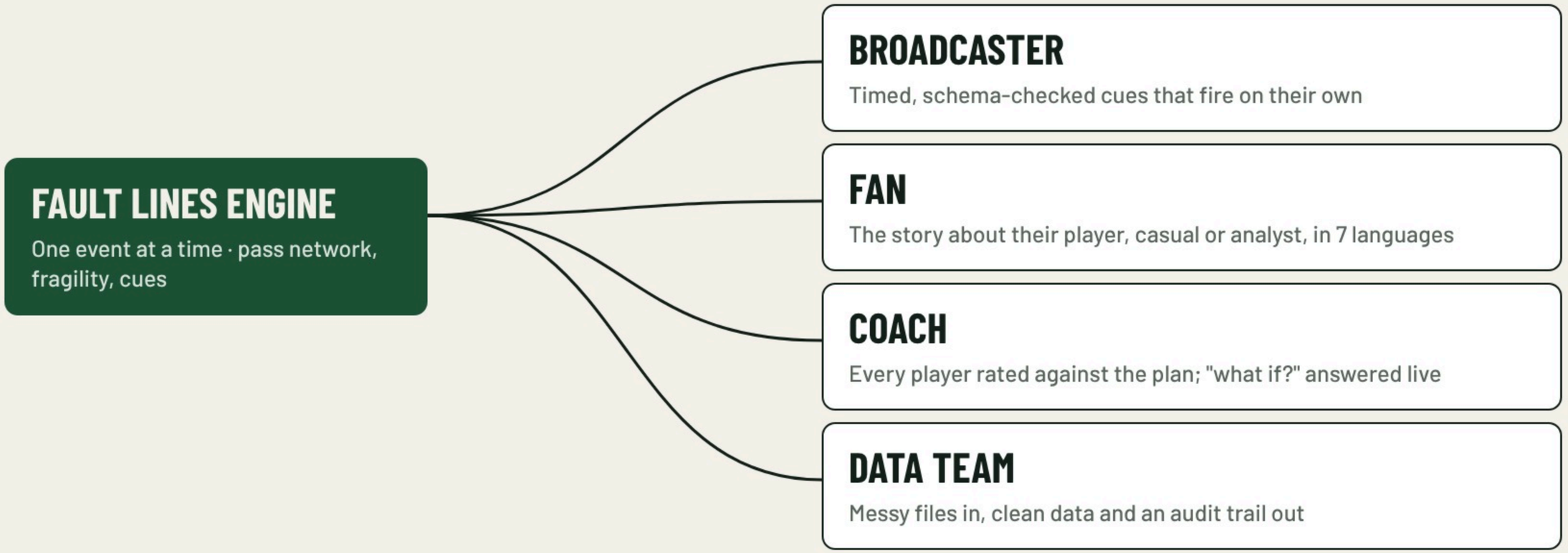
REMOVE EACH PLAYER AND MEASURE WHAT THE REST LOSE



fragility(p) = 1 - connectivity of teammates without p ÷ with p · rolling 15 min · every event

THE PRODUCT

ONE CALCULATION REACHES FOUR AUDIENCES



FOR THE BROADCASTER

THE LIVE VIEW RINGS THE FAULT LINE AS PLAY HAPPENS



Screen from the running app on a synthetic demo match with fictional clubs

RINGED IN GOLD

The fault line, and the share of links lost without them

LIVE CUES

Alerts with a match clock, streamed over Server-Sent Events

FOR THE FAN

EVERY FAN GETS THE STORY AT THEIR LEVEL, IN THEIR LANGUAGE

FAULT LINE

Luca Ferrand **39%**

Right now Northbridge FC run through Luca Ferrand. Take them out and the team loses 39% of its passing links. In the last 15 minutes they moved the ball forward 26 times (previously 31).

template

FAULT LINE

Milo Dancy **19%**

89' | Northbridge FC 2-0 Riverside Athletic. Single point of failure: Milo Dancy (RW): removing them costs 19% of passing connectivity (prev window 0%). Passing spread up 0% vs previous window; progressive passes 28 (prev 26), completion 86%. Riverside Athletic's most critical link, Yusuf Kaleo, carries 11%.

template

English Spanish French Portuguese Hindi Arabic Japanese

CASUAL OR ANALYST

Written from computed facts; a fact-check rejects any unsupported number

FOR THE COACH

THE COACH'S OWN PLAN BECOMES THE YARDSTICK



SQUAD VS GAME PLAN (plan fit 0-100: 50 = league norm for the role on the metrics your plan weights · load = share of team actions vs what the plan intends · click a player)

57 overall plan execution at 89'

56 mixed
Control the game through patient possession and short passing.

95 working
Everything goes through Luca Ferrand at the base of midfield: he dictates the tempo and connects the back line to the attack.

57 mixed
Centre-backs must play out under pressure and never give the ball away cheaply.

38 not working
Wingers create chances and counter-press to win the ball back high.

50 mixed
Theo Ashgrove is the focal point: get shots away in the box.

Player	Relevance	Plan fit	Last 15	Form	Load vs plan
Luca Ferrand · CDM	5/5	83	46	fading	1.4x (20% vs 15%)
Owen Brightwell · RCM	3/5	60	63	steady	1.5x (15% vs 10%)
Rui Calder · RCB	4/5	59	42	steady	1.1x (6% vs 5%)
Jonah Pike · LCB	4/5	58	44	steady	0.7x (4% vs 5%)
Tomas Varga · GK	2/5	54	50	steady	1.3x (1% vs 1%)
Eli Mensah · RB	3/5	53	54	steady	1.5x (5% vs 3%)
Theo Ashgrove · ST	5/5	47	56	steady	0.6x (17% vs 29%)
Kai Andersen · LCM	3/5	46	49	steady	1.0x (11% vs 10%)
Milo Dancy · RW	3/5	44	58	steady	1.2x (11% vs 9%)
Rafa Quinlan · LW	3/5	40	53	steady	1.0x (9% vs 9%)
Sami Okafor · LB	3/5	37	50	steady	0.6x (2% vs 3%)

Luca Ferrand (CDM): Everything goes through Luca Ferrand at the base of midfield: he dictates the tempo and connects the back line to the attack.

Strengths

- share of teammates' passing connectivity that runs through them 20% vs role norm 2%
- completed passes made and received: 168 vs 73.8 expected for the role by now

Judged on (weights from role + plan)

- network share ×8 · involvement ×6 · completion ×5 · press resistance ×3 · recoveries ×2 · progressive ×1

PLAN FIT, 0 TO 100

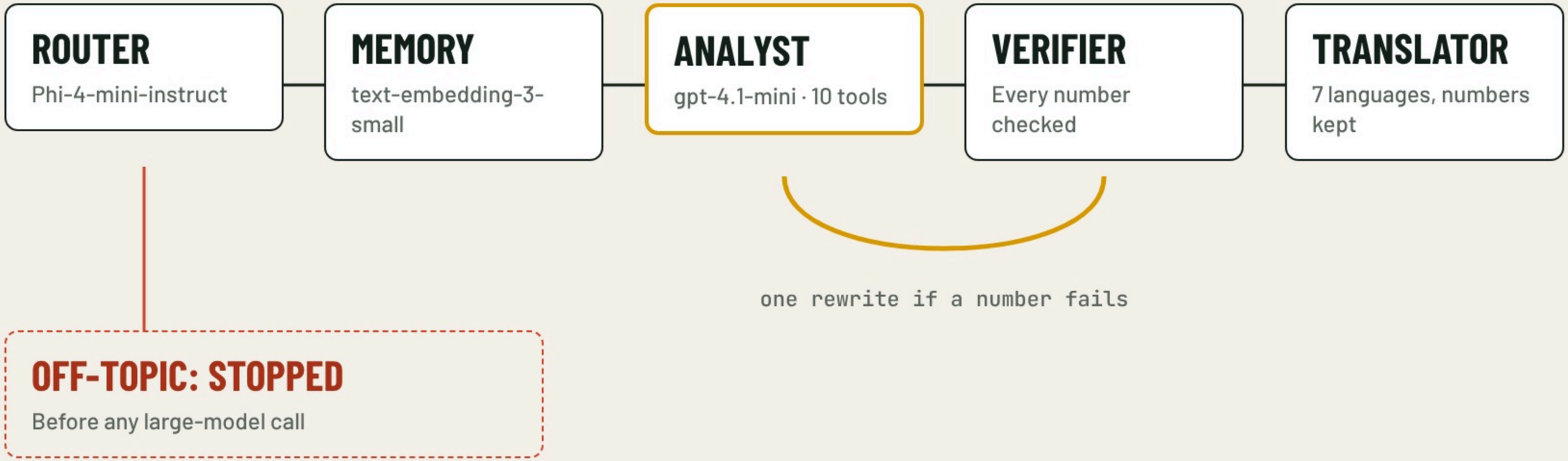
Each player judged on the metrics the coach's plan weights

FLAGS

Overloaded, fading or off plan, against what the plan intends

ASK THE ANALYST

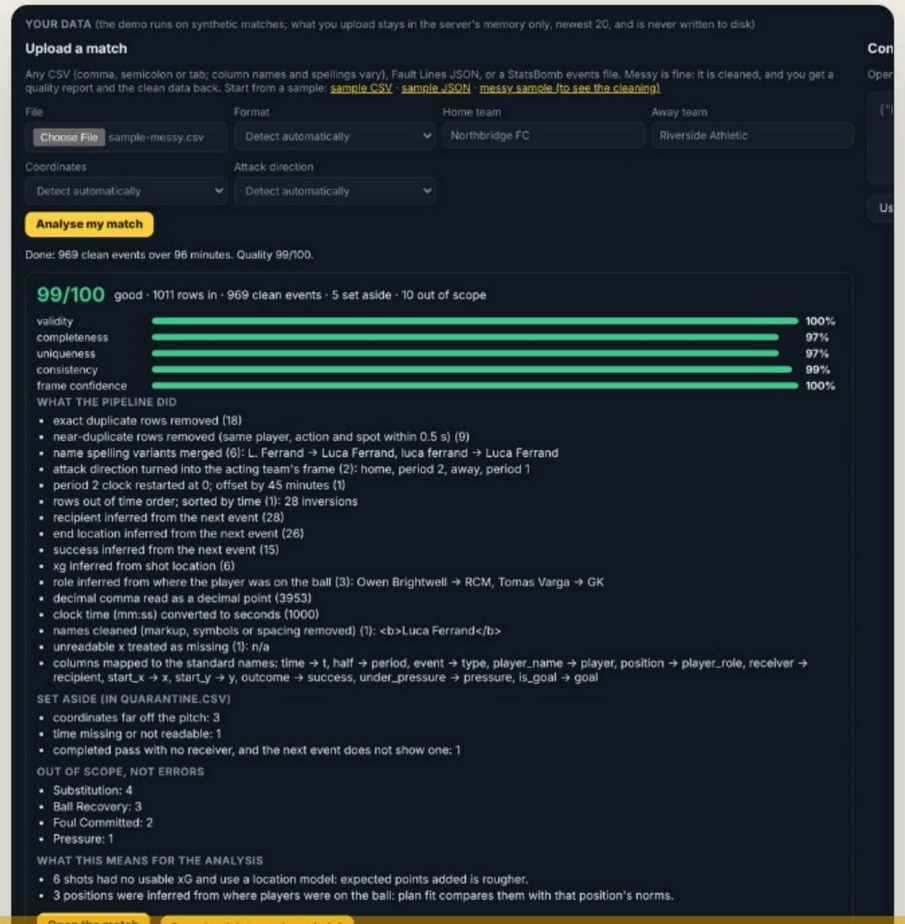
FIVE AGENTS ANSWER THE COACH, AND NONE CAN INVENT A NUMBER



Microsoft Agent Framework · models on Microsoft Foundry · tools: what-if, win odds, plan, zones

FOR THE DATA TEAM

MESSY FILES COME OUT CLEAN, WITH AN AUDIT TRAIL



QUALITY ON THE PROJECT'S MESSY TEST FILE

99

out of 100 · 1,011 rows in, 969 clean events out

EVERY FIX LISTED

CSV in almost any layout, JSON or StatsBomb; inferred values flagged; uploads never written to disk

PROOF

IT WAS TESTED BEFORE IT WAS DEMOED

6,130
real results: trained on 4,940, tested on 1,190 later ones

17/17
held-out realism checks pass

96%
of 26 coach questions answered with the right tools

4.8/5
answer quality across those questions

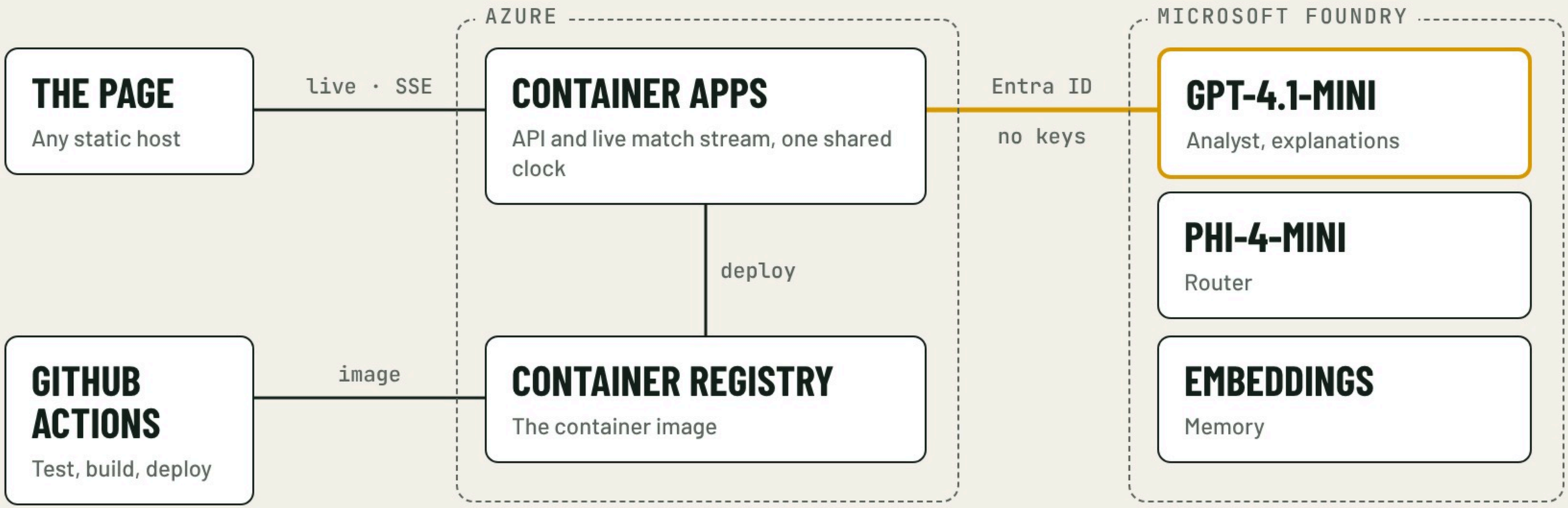
0
unchecked substitution advice

100%
of events recovered from deliberately noisy files

Sources: reports/validation.md, reports/evaluation.md · pass bars fixed before the first run · demo matches are synthetic

HOW IT'S BUILT

IT RUNS KEYLESS ON AZURE, WITH AGENTS ON MICROSOFT FOUNDRY



TRY IT

EVERY TEAM HAS A FAULT LINE.

NOW EVERYONE WATCHING CAN SEE IT.

LIVE DEMO	harshul27.github.io/faultlines
CODE	github.com/harshul27/faultlines
FILM	1:44 demo, original music

